Perturb-Seq Reanalysis Pipeline

FASTQ-to-H5AD processing for the Perturbation Catalogue

Kirill Tsukanov
Senior Full Stack Developer & Data Engineer
ktsukanov@ebi.ac.uk

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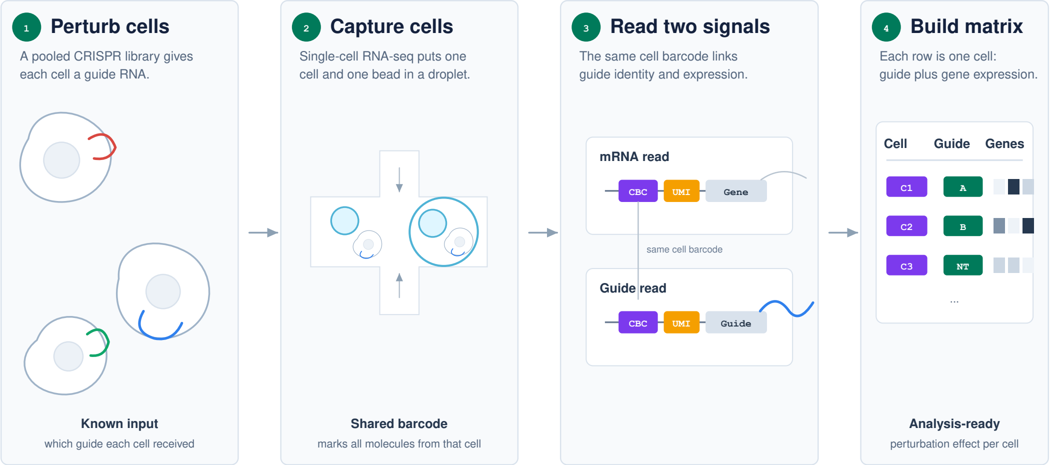
What is Perturb-Seq?



The Problem: Author H5ADs Are Not Comparable Enough

What we receive

- Processed H5AD files made for individual papers.
- Different references, gene symbols, filtering, normalization, and guide annotations.
- Metadata fields that use study-specific names and assumptions.

Why it matters

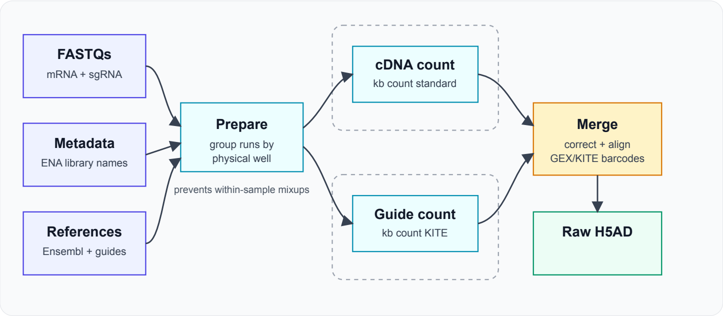
- Cross-study comparison inherits author-specific processing choices.
- Aggregating datasets can amplify processing batch effects.
- Catalogue users expect one clear schema, one count definition, and one perturbation label definition.

Decision: reprocess from FASTQ with one reference, one expression workflow, one guide workflow, and one QC policy.

Constraints and Data Issues

Data volume	The five completed datasets use about 19 TB of FASTQs; the largest single dataset is about 11 TB.
Compute budget	Storage and cluster time are adequate but not unlimited, so the pipeline must be fast enough to rerun.
Cell Ranger	Useful, but too slow for repeated catalogue-scale reprocessing of these datasets.
FASTQ source	ENA FASTQs did not include the required technical barcode/UMI reads for these runs; the current workaround uses SRA downloads.
ENA metadata	Sample names, lanes, SRRs, and library names are not consistent across studies and need dataset-specific grouping.
10x details	Expression and guide libraries can use related but non-identical barcodes; the pipeline applies the required KITE barcode correction.

Solution: Fully Automated FASTQ-to-H5AD Pipeline



- One Nextflow run builds references, groups FASTQs by physical well, quantifies expression and guides, aligns barcodes, and writes a single H5AD.
- Per-sample guide diagnostics are written before final merge, so failures are visible before downstream analysis.

Why kallisto+bustools?

- **Fast enough to rerun:** catalogue processing needs iteration, not a one-off run.
- **Open and scriptable:** the same workflow can run in Nextflow with containerized dependencies.
- **Two modalities:** standard count mode handles expression; KITE mode handles guide barcodes.
- **Works within budget:** lower wall time and storage pressure make full reprocessing practical.
- **Comparable output:** QC shows high agreement with author baselines where direct comparison is meaningful.

Guide Calling

- KITE produces a guide UMI matrix aligned to expression barcodes.
- A Gaussian-Poisson mixture separates low-count background from true guide signal.
- The decision threshold is the crossover where signal is more likely than background.
- Probe calls are collapsed to gene-level outcomes: control, single gene, multi-gene, mixed gene/control, or unassigned.
- Valid single-gene analysis uses exactly one gene-targeting call and no control guide call.

Dataset group	Threshold
Nadig HepG2	8 UMI
Nadig Jurkat	8 UMI
K562 Essential	8 UMI
K562 Genome-wide	8 UMI
RPE1 Essential	9 UMI

Thresholds come from the fitted mixture model for each dataset.

Processed Datasets

Dataset	FASTQ TB	Cells	Single-gene	Control	Multi-gene	No guide
Nadig 2025 Jurkat	2.0	588,498	214,092	10,629	289,965	58,565
Nadig 2025 HepG2	2.6	501,121	129,909	4,835	333,994	21,803
Replogle 2022 K562 Essential	1.7	544,799	274,155	11,048	191,241	61,211
Replogle 2022 K562 Genome-wide	11.0	1,314,345	613,755	29,463	545,514	101,360
Replogle 2022 RPE1 Essential	1.9	509,962	206,740	11,855	197,064	79,695
Total	**19.2**	**3,458,725**	**1,438,651**	**67,830**	**1,557,778**	**322,634**

Counts are after expression QC and Gaussian-Poisson guide calling. Mixed gene/control cells are kept separately and are not counted as valid single-gene cells.

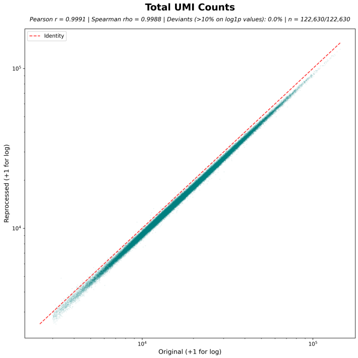
Baseline Comparison: Sanity Check, Not a Target

- The author H5AD is used to catch major processing errors.
- Exact reproduction is not the goal; it would preserve the heterogeneous processing choices we are removing.

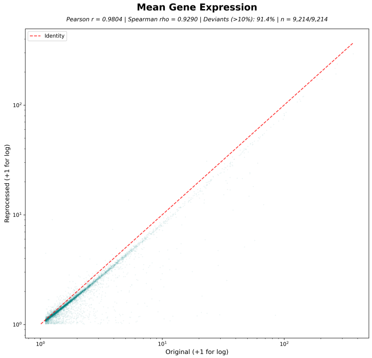
Dataset	Shared cells	Shared genes	UMI r	Mean gene r	Guide match
Nadig HepG2	122,630 (87.7%)	95.7%	0.9991	0.9804	99.92%
Nadig Jurkat	214,268 (87.5%)	95.6%	0.9992	0.9179	99.92%
K562 Essential	260,276 (91.1%)	95.4%	0.9989	0.9506	99.76%
K562 Genome-wide	596,639 (51.4%)	95.7%	0.9820	0.9583	99.87%
RPE1 Essential	210,649 (90.8%)	96.4%	0.9986	0.9021	99.91%

Replogle author matrices are signed/transformed, so cell-wise expression-vector and dropout metrics are not used as headline evidence.

Expression Check: Major Structure Is Preserved



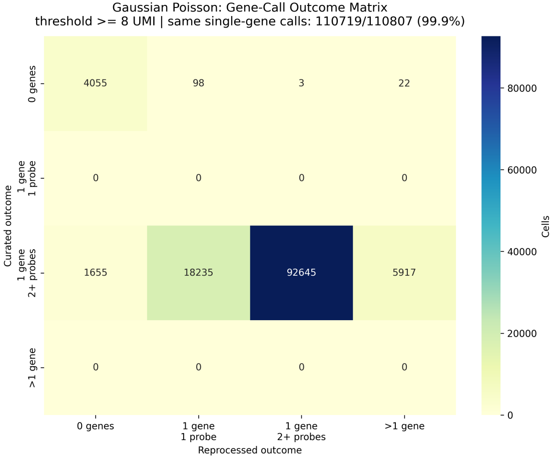
Each point is one shared cell. For large datasets, points and correlations are computed on a deterministic sample.



Each point is one shared gene.

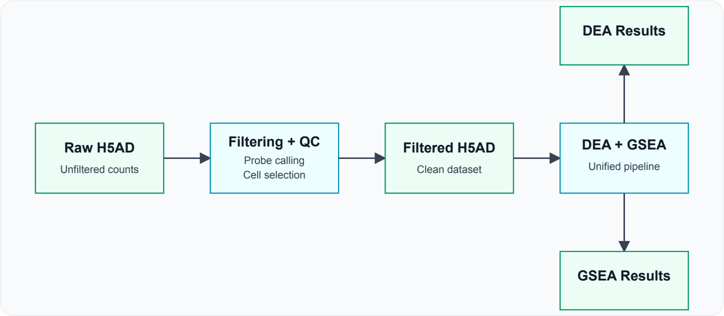
Nadig HepG2 is a raw-count baseline: shared-cell total UMIs correlate at $r = 0.9991$ and shared-gene mean expression at $r = 0.9804$.

Guide Calls Agree With the Author Labels



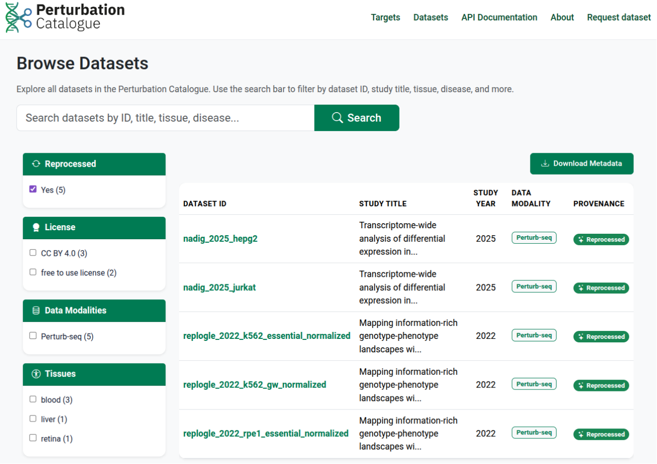
- HepG2: 110,719 / 110,807 shared valid single-gene cells have the same target gene.
- Across all five datasets, valid single-gene match is 99.76–99.92%.
- Multi-gene and no-guide cells are kept as explicit categories rather than forced into a single target.

Secondary Analysis



- Differential expression compares each perturbation with control cells using Scanpy Wilcoxon.
- GSEA uses preranked gene lists and writes DEA/GSEA Parquet outputs for Catalogue ingestion.

In Production Now



- Five reprocessed datasets are visible in the Catalogue.
- The dataset browser exposes a `Reprocessed` filter.
- Users can download metadata and open the dataset pages now.

Next Steps

- Add more evaluation checks: cross-dataset QC summaries, metadata checks, and downstream-signal checks.
- Package the FASTQ-to-H5AD pipeline as a separate reusable project with versioned containers and a small test dataset.
- Add support for more data types and chemistries, starting with 10x v2 and cell surface protein data.
- Continue ingesting more Perturb-Seq datasets once FASTQ grouping and guide metadata are mapped.